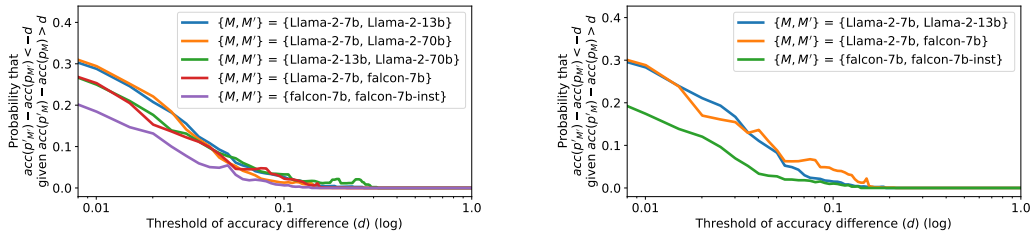
B.2 Additional Results for Section 4.2

Table 3: Ratio of prompt format pairs (p_1, p_2) such that if p_1 is worse than p_2 using model M_1, then the same trend holds for M_2.

Model 1 (M_1)	Model 2 (M_2)	Performance Relative Ordering Preservation
Llama-2-7b	Llama-2-13b	57.46%
Llama-2-7b	Falcon-2-7b	55.91%
Falcon-7b	Falcon-7b-Inst	61.11%



(a) Option ranking metric.

(b) Exact prefix matching metric.

Figure 12: Probability of a prompt p being worse than p' by at least d points using model M', given that prompt p was *better* than prompt p' when using model M.

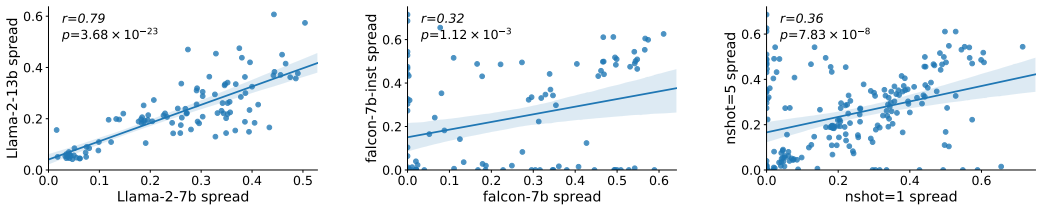
Formats are not inherently good or bad. Table 3 shows that if format p_1 has lower performance than format p_2 under model M, there is < 0.62 probability that this trend would hold under another model M' (random chance is 0.5). This weak relative order preservation suggests that prompt format performance in a model may not be extrapolated to a different model, or in other words, that there are no inherently good or bad formats. This finding is further supported by Figure 12, which shows that findings of a format being better or worse than another are often inconsistent across models.

Experiments with exact prefix matching accuracy. Here we show results with using exact prefix matching to compute accuracy. Often, failures in prefix matching are associated with degeneration, i.e., cases where the model does not answer any of the valid options, motivating the use of ranking accuracy. Degeneration makes models (specially smaller models) more unlikely to have high accuracy out of the box. As seen in Figure 6, prefix matching is linked to having higher changes when performing atomic changes. Moreover, exact prefix matching can lead to lower performance as generation is less constrained (see Figure 16). Table 4 shows examples of atomic changes yielding large accuracy changes with exact prefix matching metric.

Figure 13c shows spread remains regardless of model size increase, architecture change, or number of few-shot examples also when using exact prefix matching as accuracy metric. In line with the results shown for probability ranking in Section 4.2, Figure 15 shows that the probability of reversing performance trends between two models just by changing prompt remains high when using exact prefix matching as metric. Strikingly, spread is significantly higher than in the probability ranking setting (see Figure 14), with median spread ranging from 12 to 28 accuracy points depending on the model used. This further motivates the need for running FormatSpread when benchmarking models with this accuracy metric. This increased spread may be partly due to degeneration, as we will detail next.

Degeneration. Sometimes when a model does not generate the correct answer with exact prefix matching, it also does not generate a valid response, i.e. it degenerates. We will now quantify this phenomenon using 53 SuperNaturalInstructions classification and multiple choice tasks.

Given a model, a task, and a format, let the *centered mass* be the ratio of examples where the model's output matched with any valid option (regardless of correctness). Table 5 shows that the correlation between accuracy and centered mass is moderate or high depending on the model. This suggests that very often when a model does not return a valid answer, it does not return any valid answer



(a) Llama-2-7B vs. 13B (b) Falcon-7B vs. 7B-Instruct (c) 1- vs. 5-shot (same task, model)

Figure 13: Spread comparison between evaluating the same task under different models or n-shots using exact prefix matching as accuracy metric.

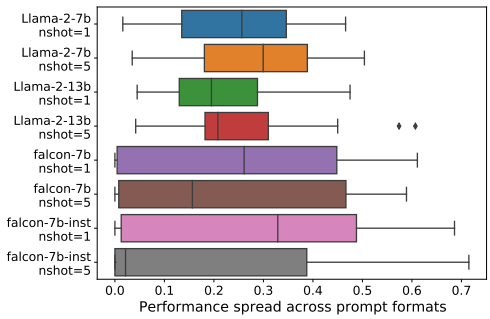


Figure 14: Spread across models and n-shots. Exact prefix matching metric.

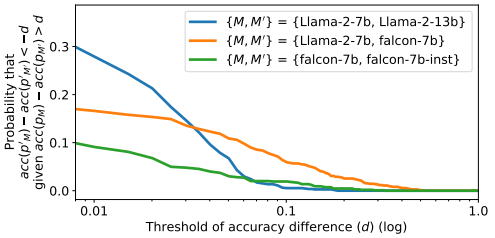
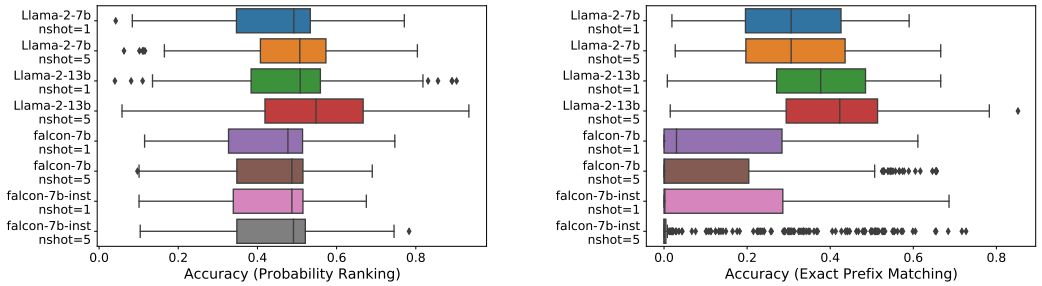


Figure 15: Probability that model M performs *worse* than M' by at least d when using format p', given that M performed *better* than M' by at least d using format p. 53 tasks, 1- and 5-shot. Exact prefix matching metric.



(a) Accuracy boxplot for the selected 53 Super Natural-Instructions tasks, option ranking metric.

(b) Accuracy boxplot selected 53 Super Natural-Instructions tasks, exact prefix matching metric.

Figure 16: Accuracy metric used can strongly impact final performance. 53 Super Natural-Instructions tasks shown. Ranking accuracy yields higher accuracies overall.

at all. This is especially true for Falcon models, where we observe an almost perfect correlation between accuracy and centered mass. In conclusion, prompt format chosen often do not solely affect accuracy, but they also affect the frequency in which a model is actually able to perform a task. This will especially affect tasks for which there are no alternative metrics. Further research may focus specifically on targeting features that cause degeneration.

Experiments with Instruction Induction tasks. All experiments thus far focused solely on classification tasks. We will now focus on tasks that require generating (short) text, and cannot be framed as classification tasks. We selected 10 tasks from Instruction Induction (Honovich et al., 2023) that require generating a unique, valid string to be considered a correct response. Examples include identifying the second letter of a word, adding numbers, or answering a synonym to a given word. Instruction Induction tasks also show a wide range of difficulty, resulting in varied settings to be

Table 4: Examples of atomic changes' impact on accuracy using prefix matching (probability ranking shown in Table 2). {} represents a text field; p_2 yields higher accuracy than p_1 for all tasks.

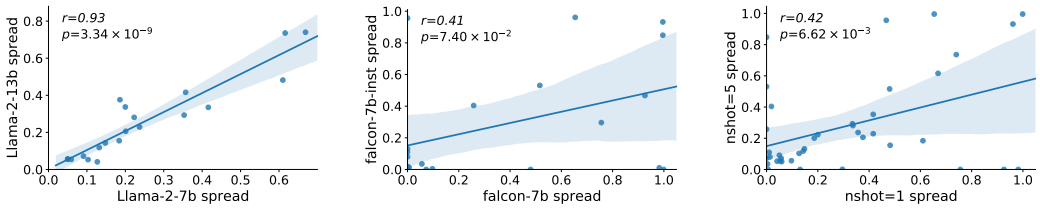
Task Id	Prompt Format 1 (p_1)	Prompt Format 2 (p_2)	Acc p_1	Acc p_2	Diff.
task213	`Title: {} Sentence<A>: {} || Sentence<B>: {} || Sentence<C>: {} || Sentence<D>: {} Choices: \n <i>::: {} \n <ii>::: {} Answer: {}'`	`Title::{} Sentence<A>::{} || Sentence<B>::{} || Sentence<C>::{} || Sentence<D>::{} Choices::\n <i>::: {} \n <ii>::: {} Answer::{}'`	0.113	0.475	0.362
task296	`Sentence I ) : {} \nSentence II ) : {} \nSentence III ) : {} \nSentence IV ) : {} \nSentence V ) : {} \nSentence VI ) : {} \nSentence VII ) : {} \nSentence VIII ) : {} \nSentence IX ) : {} \nSentence X ) : {} , I. : {} , II. : {} , Answer: {}'`	`Sentence I ) : {} \nSentence II ) : {} \nSentence III ) : {} \nSentence IV ) : {} \nSentence V ) : {} \nSentence VI ) : {} \nSentence VII ) : {} \nSentence VIII ) : {} \nSentence IX ) : {} \nSentence X ) : {} , 1. : {} , 2. : {} , Answer: {}'`	0.201	0.522	0.321
task905	`Tweet::: {}; \nLabel::: {}; \nAnswer::: {}'`	`Tweet::{}; \nLabel::{}; \nAnswer::{}'`	0.252	0.559	0.307
task317	`Passage:: {} \nAnswer:: {}'`	`Passage::{} \nAnswer::{}'`	0.245	0.546	0.301
task280	`passage {}\n answer {}'`	`passage:{}\n answer:{}'`	0.332	0.612	0.28
task050	`SENTENCE - {} \nQUESTION - {} \nANSWER - {}'`	`SENTENCE\n\t{} \nQUESTION\n\t{} \nANSWER\n\t{}'`	0.244	0.504	0.26
task070	`Beginning - {}\nMiddle [I]{} , Middle [II]{}\nEnding - {}\nAnswer - {}'`	`Beginning - {}\nMiddle I){} , Middle II){}\nEnding - {}\nAnswer - {}'`	0.143	0.3	0.157

Table 5: Correlation between accuracy using exact prefix matching and the centered mass (the opposite of degeneration). 53 tasks, 10 formats each, evaluated on 1000 samples.

Model	n-shot	correlation between accuracy & and centered mass	p-value
Llama-2-7b	1	0.702	5.1E-77
Llama-2-7b	5	0.762	4.9E-98
Llama-2-13b	1	0.639	5.8E-61
Llama-2-13b	5	0.662	9.2E-67
falcon-7b	1	0.936	7.1E-233
falcon-7b	5	0.933	8.4E-228
falcon-7b-instruct	1	0.962	3.6E-289
falcon-7b-instruct	5	0.958	5.5E-277

analyzed (see Figure 18b). Given that the collection does not contain human-generated formats, we applied a simple '`Input: {}\n Output: {}`' format. Results for 1-shot and 5-shot settings show spread is still high across models and n-shot choices (see Figure 17).

Tasks are: `antonyms, diff, first_word_letter, larger_animal, letters_list, num_to_verbal, second_word_letter, singular_to_plural, sum, synonyms.`



(a) Llama-2-7B vs. 13B (b) Falcon-7B vs. 7B-Instruct (c) 1- vs. 5-shot (same task, model)

Figure 17: Spread comparison between evaluating the same task under different models or n-shots for Instruction Induction tasks. Exact prefix matching used as accuracy metric.